Modeling and Validation of a Soccer Ball's Three-Dimensional Flight to the Goal

Problem B

Abstract

This study predicts a football's goal-plane crossing and its uncertainty from an initial position and two subsequent observations. Four models are compared: gravity alone, gravity with drag, constant resultant force, and gravity with drag and Magnus force. The model incorporating gravity, drag, and Magnus force uses a constant drag coefficient and a linear lift relation. Initial velocity, spin direction, and aerodynamic parameters are inferred from early positions, followed by numerical trajectory integration.

Subsequently, two placement shots and one power shot are selected as representative cases spanning different kick intensities and initial conditions. Predictions are evaluated against subsequent measured positions; the model incorporating gravity, drag, and Magnus force yields crossing errors of 0.360, 0.146, and 0.113 m, respectively. For the given problem, the reference crossing is (11, 2.616, 1.953) m, reached at approximately 0.399 s.

Sensitivity analysis examines initial motion, aerodynamic parameters, and air density. A 1° change in launch direction shifts the crossing by approximately 0.20 m, while a 5% speed change alters arrival time by up to 0.021 s. Spin magnitude and the lift parameter jointly determine deflection through their product. Different aerodynamic and initial-motion combinations reproduce the early positions within a 2 mm coordinate tolerance, while their crossings deviate from the baseline trajectory by up to 0.1074 m within the specified aerodynamic parameter ranges. Next, independent uniform perturbations of nine parameters generate 100,000 Monte Carlo trajectories, defining a crossing distribution and a joint 95% region; 95.772% clear the goal opening.

Keywords: penalty trajectory; aerodynamic drag; Magnus effect; parameter estimation; uncertainty analysis

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1 Introduction

1.1 Problem Background

A penalty kick is a direct contest between a designated kicker and the opposing goalkeeper. The kicker strikes the ball from the penalty mark towards the goal. In a typical professional penalty, the ball then flies freely through the air towards the goal, while the goalkeeper must choose where and when to move. An effective shot depends on speed and placement, as well as the kicker's ability to conceal its intended direction.

The importance of penalties to match outcomes has motivated research into shooting and saving, particularly the ball's motion after it leaves the foot. Topics include curved trajectories and aerodynamic forces[3], surface structure[5, 4], and trajectory measurements and aerodynamic-parameter estimation[9]. These studies connect kicking technique with the physics of ball flight and help explain the motion of a penalty shot.

1.2 Problem Restatement

Problem B concerns the prediction of a penalty trajectory from its early flight positions. The goal is 11 m from the penalty mark, 7.32 m wide and 2.44 m high. The task is to develop and test a general method that uses the ball-centre coordinates at 0.05 s and 0.10 s after the kick to predict its most likely entry point and estimate the uncertainty of that prediction.

The preceding discussion suggests three broad stages of the ball's motion: **contact with the kicker**, **free airborne flight** and **motion after the shot**. We examine the free flight after ball–foot contact has ended and before the ball enters the goal; the effect of contact is represented by the initial velocity and spin. The forces and governing equations are discussed further in Section 3. Initial-state inference from early positions, subsequent trajectory prediction and assessment of those predictions are discussed further in Section 4. Section 5 uses the specified example to examine parameter sensitivity and the distribution of goal-plane crossings under stated perturbations, connecting the predicted location with its uncertainty. The technical workflow is summarized in Figure 1.

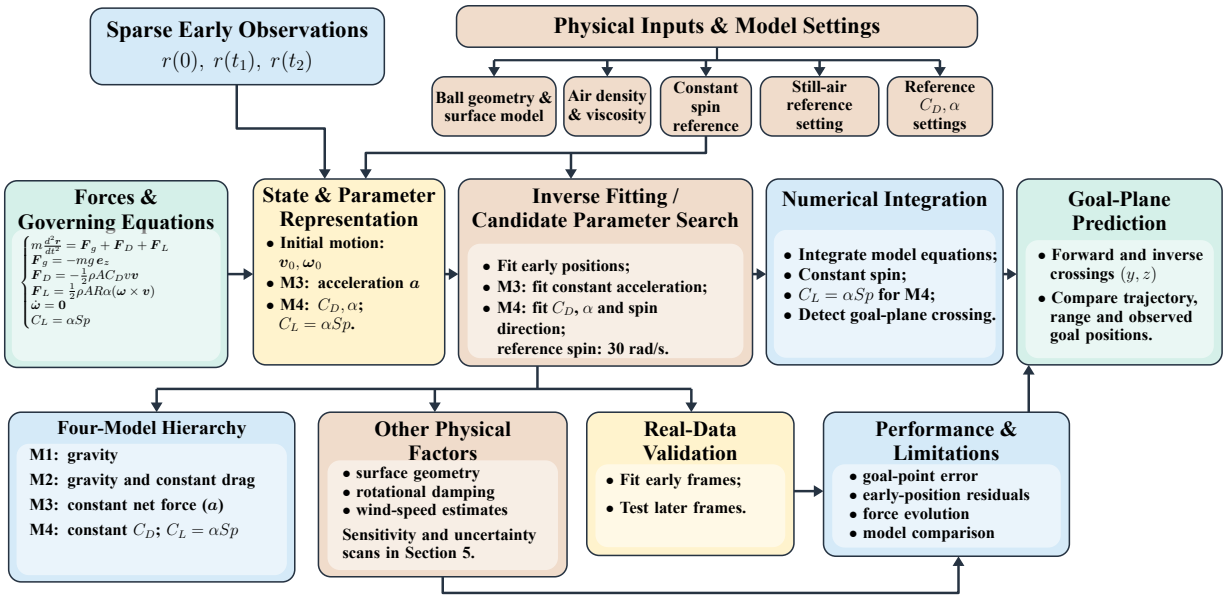


Figure 1: Three-force modeling, inverse prediction and real-penalty validation.

2 Preparation of the Models

2.1 Assumptions

1. **Free flight.** At the start of the motion considered, the ball has left the kicker's foot; contact is represented by the initial velocity and spin. We assume a proficient kick, with the ball remaining airborne without contacting the ground or another object before reaching the goal plane.

2. **Rigid spherical ball.** The ball remains spherical and its mass is uniformly distributed. Seams and asymmetric surface geometry are discussed further in Section 6.1.

3. **Uniform gravity and air properties.** The magnitude and direction of gravitational acceleration remain constant over the flight, and air density and viscosity remain uniform.

4. **Still air.** The surrounding air is treated as stationary. The effects of wind on relative airflow and aerodynamic force are discussed further in Section 6.3.

5. **Constant spin.** Aerodynamic torque is neglected, so the spin vector remains constant during flight. Rotational damping is discussed further in Section 6.2.

2.2 Notations

The symbols used below and their physical meanings are summarized in Table 1.

Table 1: Principal quantities and competition reference values.

Symbol	Meaning	Value or unit
$\boldsymbol{r}, \boldsymbol{v}$	Ball-centre position and velocity	m; m/s
$\boldsymbol{\omega}, \omega$	Spin vector and its magnitude	rad/s
m, R, A	Mass, radius and frontal area	0.43 kg; 0.11 m; πR^2
g	Gravitational acceleration	9.81 m/s^2
ρ	Air density	1.2 kg/m^3
$\nu, \mu = \rho\nu$	Kinematic and dynamic viscosity	1.50×10^{-5} m^2/s; Pa s
Re, Sp	Reynolds number and spin parameter	dimensionless
C_D, C_L, α	Drag, lift and linear-spin parameters	dimensionless
$\lambda_D, \lambda_\alpha$	Coefficient ratios to reference values	dimensionless
$\theta, \boldsymbol{e}_L$	Spin–airflow angle and Magnus direction	rad; unit vector
q, v	Dynamic pressure and ball speed	Pa; m/s
$\boldsymbol{Q}, I_b$	Damping torque and moment of inertia	N m; kg m^2

We follow Maponi and Piombin [3] for the ball parameters and Asai et al. [5] for the reference air properties, with dynamic viscosity $\mu = \rho\nu$.

3 Model Establishment

3.1 Geometric Relations

As shown in Figure 2, the local origin is the penalty mark, with x towards the goal, y horizontal and lateral, and z vertically upward. The goal plane is $x = 11$ m and the opening is $-3.66 \le y \le 3.66$ m, $0 \le z \le 2.44$ m.

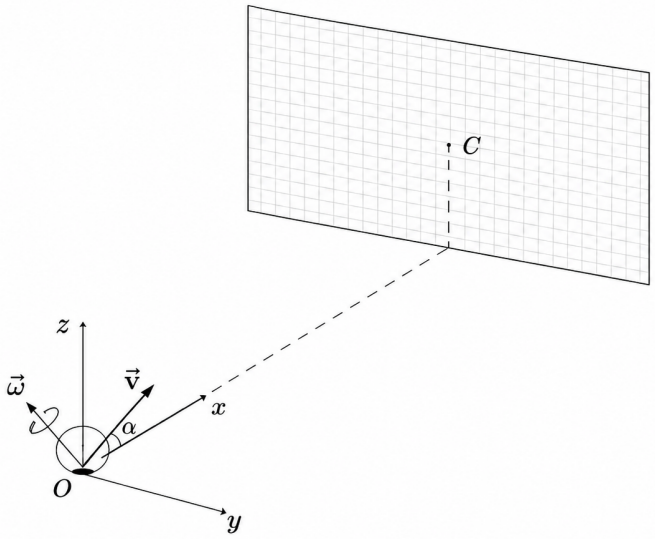


Figure 2: Penalty-kick geometry in local coordinates; O is the nominal penalty mark and C an illustrative goal-plane point.

The entire ball clears the opening when its centre satisfies $|y| \leq 3.66 - R$ and $R \leq z \leq 2.44 - R$.

3.2 Force Analysis

As shown in Figure 3, we consider gravity, buoyancy, drag and Magnus force in turn.

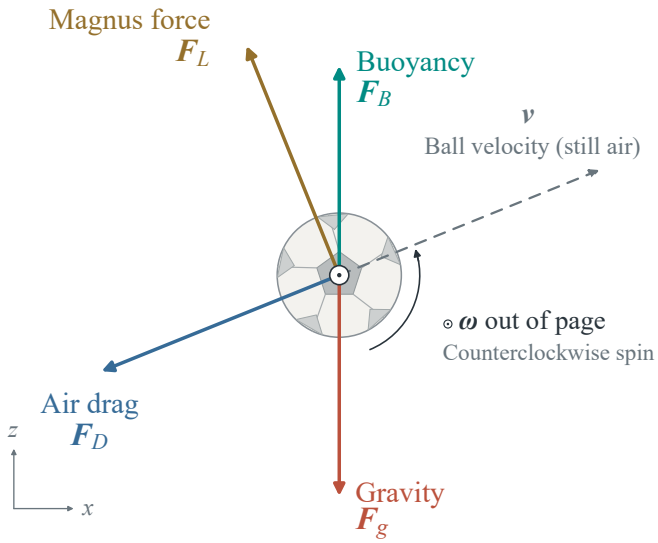


Figure 3: Schematic of gravity, buoyancy, drag and Magnus force on the football.

3.2.1 Gravity

Let $\boldsymbol{e}_z$ be the upward vertical unit vector. For mass m in the uniform field of gravitational acceleration magnitude g,

$$\boldsymbol{F}_g = -mg\boldsymbol{e}_z, \tag{1}$$

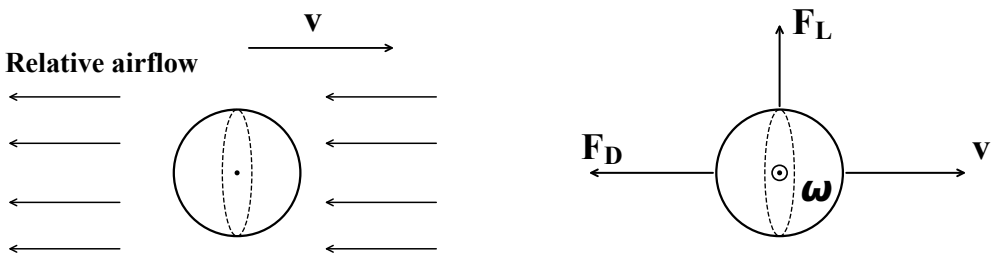
By Archimedes' principle, the displaced air also exerts an upward buoyancy force $\boldsymbol{F}_B$. For ball volume $V_b = \frac{4}{3}\pi R^3$ and air density ρ,

$$\boldsymbol{F}_B = \rho V_b g\, \boldsymbol{e}_z = \frac{4}{3}\pi R^3 \rho g\, \boldsymbol{e}_z. \tag{2}$$

From these expressions, the combined vertical force is $\boldsymbol{F}_g + \boldsymbol{F}_B = -(m - \rho V_b)g\boldsymbol{e}_z$. Buoyancy is 1.56% of weight for the reference parameters and is therefore omitted as a small correction in the subsequent calculations.

3.2.2 Aerodynamic Drag

Let $\boldsymbol{v}$ be the ball-centre velocity. Since the air is assumed stationary, the incident airflow in the ball frame has velocity $-\boldsymbol{v}$. As shown in Figure 4, the airflow slows near the forward face of the ball.



(a) Airflow in the ball frame **(b) Example: spin out of the page**

Figure 4: Schematic of relative airflow and force directions.

Let p_∞ be the incident static pressure and p_s the pressure at the forward stagnation point. Bernoulli's relation along an idealized incident streamline is

$$p_\infty + \frac{1}{2}\rho v^2 = p_s.$$

The pressure rise defines the dynamic pressure q:

$$q = p_s - p_\infty = \frac{1}{2}\rho v^2.$$

Following Goff and Carré [9], aerodynamic drag depends on air density, relative speed, frontal area and the ball's flow and surface characteristics. The latter enter through the drag coefficient C_D, so the drag magnitude is

$$F_D = C_D q A.$$

Since drag opposes the relative motion, substitution of the dynamic pressure leads to

$$\boldsymbol{F}_D = -\frac{1}{2}\rho A C_D v \boldsymbol{v}. \tag{3}$$

Here $A = \pi R^2$ is the frontal area and ρ is air density. We approximate C_D as constant during the ball's flight.

3.2.3 Magnus Lift

A spinning football changes the surrounding airflow and experiences a transverse Magnus force. As shown in Figure 5, rotation makes the relative flow asymmetric. Let $\boldsymbol{\omega}$ be the angular-velocity vector.

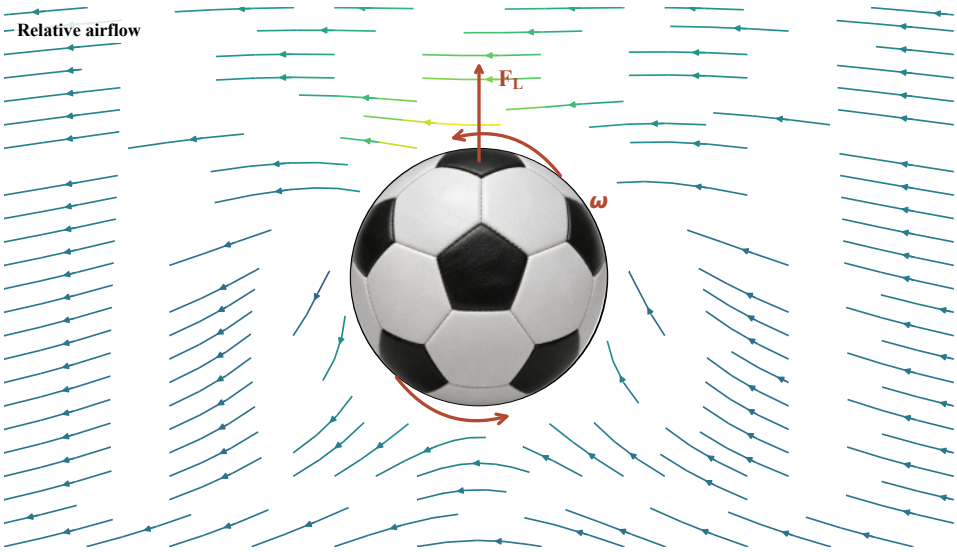


Figure 5: Idealized flow around a spinning football.

Following Goff and Carré [9], transverse lift is expressed in terms of dynamic pressure and frontal area. When the spin axis makes angle θ with the relative velocity, the lift magnitude is

$$F_L = qAC_L \sin\theta.$$

Here C_L is the transverse lift coefficient and θ is the spin–airflow angle. We approximate the relation between lift coefficient and spin parameter as linear:

$$C_L = \alpha \frac{R\omega}{v}.$$

Here α is dimensionless. Based on Kiratidis and Leinweber [6], we approximate this coefficient as constant during the flight. Substituting the dynamic-pressure and linear-spin relations,

$$\boldsymbol{F}_L = \frac{1}{2}\rho A R \alpha (\boldsymbol{\omega} \times \boldsymbol{v}). \tag{4}$$

3.3 Dynamical Equations

Since the football is treated as a rigid body, its motion can be separated into translation of the centre of mass and rotation about that centre. Let $\boldsymbol{L}$ be angular momentum about the centre, I_b the moment of inertia, and $\boldsymbol{Q}$ the resultant external torque. Centre-of-mass translation follows Newton's second law,

$$m\dot{\boldsymbol{v}} = \sum \boldsymbol{F} \tag{5}$$

while rotation about the centre follows the angular-momentum theorem,

$$\frac{d\boldsymbol{L}}{dt} = \boldsymbol{Q}, \qquad \boldsymbol{L} = I_b\boldsymbol{\omega}. \tag{6}$$

From zero applied torque and constant spherical inertia, $I_b\dot{\boldsymbol{\omega}} = \boldsymbol{0}$, so the spin remains constant. Combining this result with the three retained forces,

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ \dot{\boldsymbol{\omega}} = \boldsymbol{0}. \end{cases} \tag{7}$$

Let $\boldsymbol{r}_0$, $\boldsymbol{v}_0$ and $\boldsymbol{\omega}_0$ be the initial centre position, velocity and angular velocity, respectively; their subscripts x, y, z denote components in the chosen axes. The initial conditions are

$$\begin{cases} \boldsymbol{r}(0) = \boldsymbol{r}_0, \\ \boldsymbol{v}(0) = (v_{x0}, v_{y0}, v_{z0}), \\ \boldsymbol{\omega}(0) = (\omega_{x0}, \omega_{y0}, \omega_{z0}). \end{cases} \tag{8}$$

4 Model Solving

This chapter develops models and an algorithm for inferring the initial motion from early ball positions and predicting the goal-plane crossing.

4.1 Solution Method

4.1.1 Model selection

Based on the force analysis in Chapter 3, we compare four models with different force assumptions:

1. **M1: gravity.** With $\boldsymbol{a} = (0, 0, -g)$, only the three components of $\boldsymbol{v}_0$ are estimated. Spin has no effect on the trajectory.

2. **M2: gravity and constant-coefficient drag.** Only weight and drag opposite to velocity are considered. The estimated parameters are the initial velocity and constant C_D, for four unknowns in total.

3. **M3: constant resultant force.** All forces are represented by a constant resultant acceleration $\boldsymbol{a}$. The three components of $\boldsymbol{v}_0$ and three of $\boldsymbol{a}$ form six unknowns. Gravity is included in this resultant.

4. **M4: gravity, drag and Magnus force.** The force relations of Section 3 determine the motion. We estimate three initial velocity components, two spin-direction angles, C_D and the linear lift parameter α: seven unknowns.

4.1.2 Inverse method

For measured positions $\boldsymbol{r}_i$ at times t_i ($i = 0, 1, 2$), let $\widehat{\boldsymbol{r}}(t_i; \boldsymbol{p})$ denote the prediction associated with the unknown parameter vector $\boldsymbol{p}$. We minimize the six coordinate residuals through

$$J(\boldsymbol{p}) = \sum_{i=1}^{2} \left\| \widehat{\boldsymbol{r}}(t_i; \boldsymbol{p}) - \boldsymbol{r}_i \right\|^2 .$$

The parameter bounds for the four models appear in Appendix A.

After fitting, the inferred state is integrated to the goal plane $x = 11$ m. The goal-crossing distance error is

$$E_g = \sqrt{(\widehat{y}_g - y_g)^2 + (\widehat{z}_g - z_g)^2}, \tag{9}$$

where (y_g, z_g) is the observed crossing and hats denote the prediction. Further parameter and source records appear in Appendix A.

4.1.3 Dataset

To assess the general applicability of the models, we use the Sportec Open DFL dataset described by Bassek et al. [1] and released by DFL [2], which combines match events with 25 Hz multi-camera tracking. We select the following three penalties, referred to as Cases 1, 2 and 3:

1. **Case 1:** Dejan Ljubicic's penalty in 1. FC Köln–FC Bayern München on 27 May 2023.
2. **Case 2:** Philipp Klement's penalty in Fortuna Düsseldorf–1. FC Kaiserslautern on 11 November 2022.
3. **Case 3:** Dawid Kownacki's penalty in Fortuna Düsseldorf–SSV Jahn Regensburg on 26 August 2022.

The selected starting positions are $\boldsymbol{r}_0 = (3.04, 0.55, 0.13)$ m, $(2.57, 0.85, 0.34)$ m and $(0.77, 0.08, 0.24)$ m for Case 1, Case 2 and Case 3, respectively; each defines $t = 0$. As shown in Table 2, the next two positions occur at 0.04 and 0.08 s.

Table 2: Two later input positions and observed goal crossings; coordinates in m.

Case	t (s)	(x, y, z)	Observed (y_g, z_g)
Case 1	0.04	$(4.19, 0.82, 0.15)$	$(2.7500, 0.1600)$
Case 1	0.08	$(5.25, 1.07, 0.18)$	
Case 2	0.04	$(3.64, 1.20, 0.41)$	$(3.0697, 0.4657)$
Case 2	0.08	$(4.71, 1.53, 0.47)$	
Case 3	0.04	$(1.99, 0.34, 0.46)$	$(2.9188, 1.1789)$
Case 3	0.08	$(3.18, 0.62, 0.65)$	

4.2 Case Analysis

According to Hunter et al. [11], penalty technique and ball speed affect accuracy and the goalkeeper's anticipation. Studying the kicker's technique helps interpret and predict penalty trajectories. We distinguish two common approaches:

1. **Precise placement shot.** A side-foot contact favours control of direction and accurate placement. A low trajectory can carry the ball towards a lateral target while keeping it close to the turf.

2. **Power instep shot.** An instep contact emphasizes a fast strike, shortening the goalkeeper's available response time. The launch direction can produce a higher flight towards the goal.

Reviewing the match footage, we classify Case 1 and Case 2 as placement shots and Case 3 as a power shot. As shown in Figure 6, Figures 6a and 6b illustrate the low placement and higher power-shot flight patterns, respectively.



(a) Illustrative placement shot.



(b) Illustrative power shot.

Figure 6: Photographic examples of placement and power shots.

4.2.1 Precise placement shots

As shown in Figure 7, both placement shots remain low.

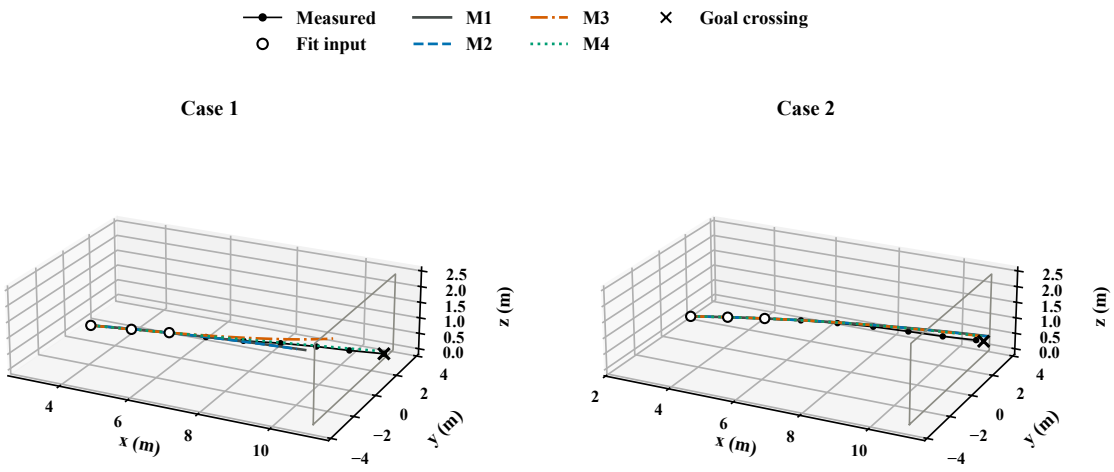


Figure 7: Measured and predicted trajectories for the two placement shots.

In M4, the initial vertical velocities are 0.55 and 1.89 m/s for the two cases. As shown in Figures 7 and 8, M1 and M2 touch the ground early in Case 1, and M3 does not reach the goal plane, while M4 stays closer to the measured flight. Comparing the four models suggests that the upward Magnus force associated with spin partly offsets gravitational fall and allows the ball to continue towards the goal along a low trajectory. In Case 2, both M3 and M4 reproduce bending in the y direction, with M4 providing the most accurate prediction. Thus, the small upward velocity keeps a placement shot low, while Magnus force plays a key role in maintaining its height.

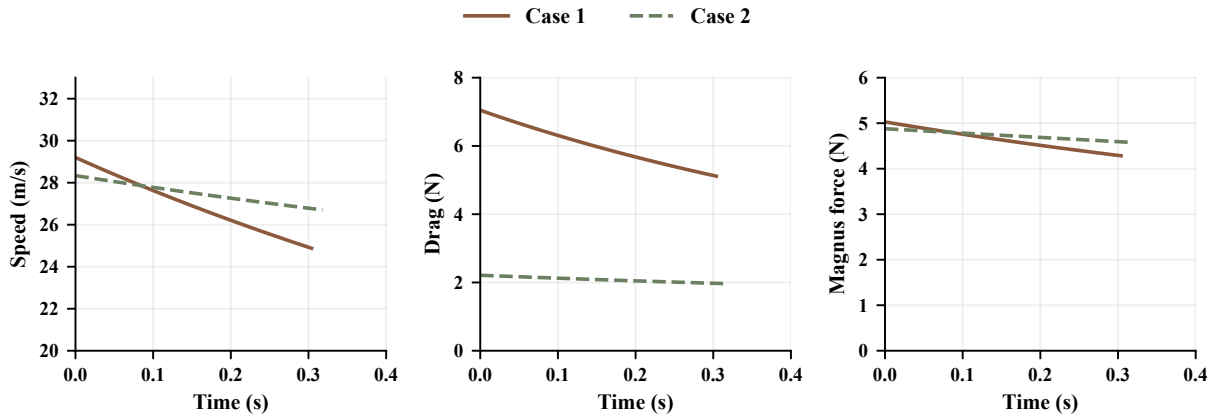


Figure 8: M4 speed, drag and Magnus-force magnitude for the two placement shots. Axis scales match Figure 10.

4.2.2 Power instep shot

As shown in Figure 9, Case 3 rises substantially higher than the two placement shots.

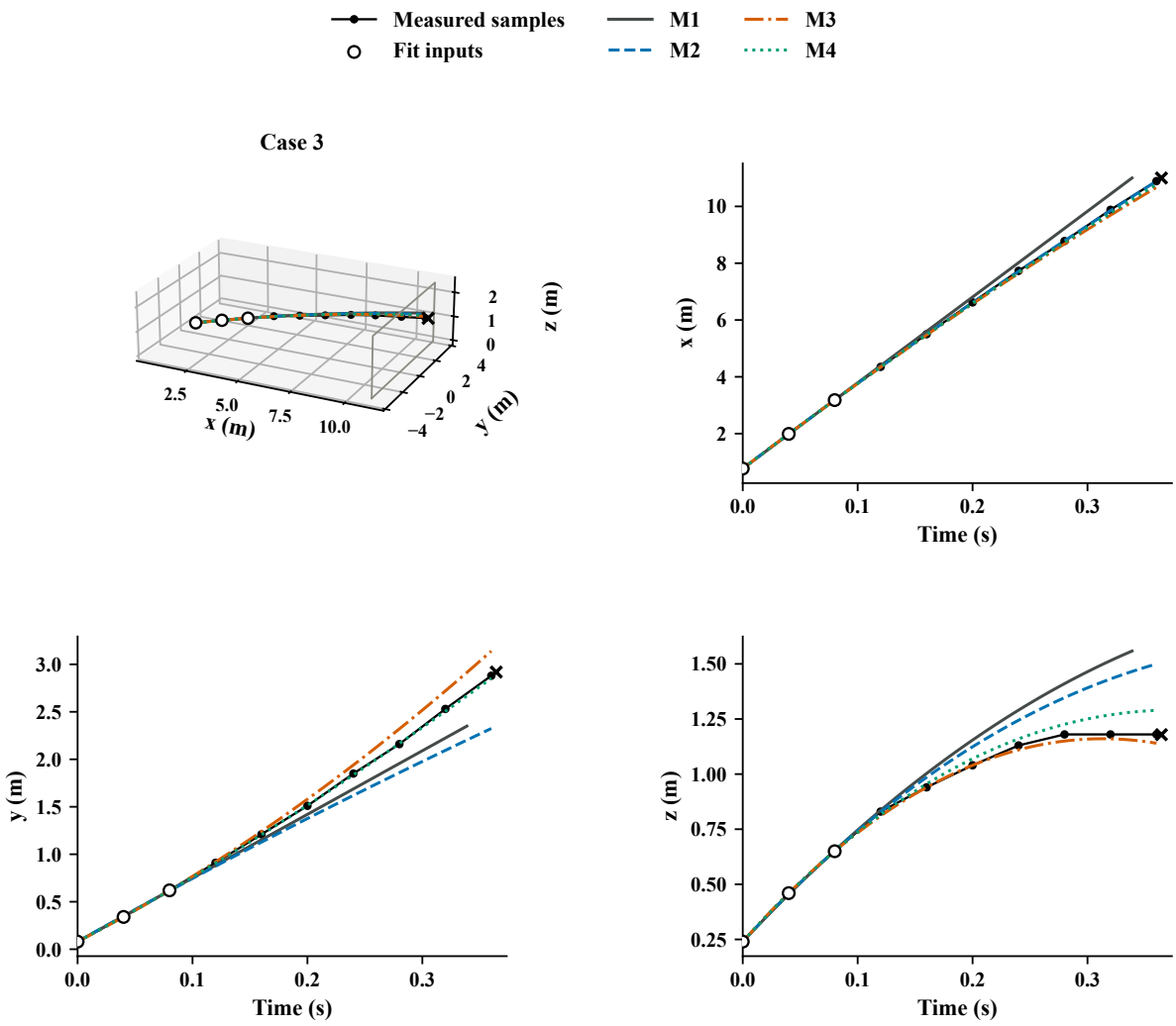


Figure 9: Case 3: spatial trajectory, $x(t)$, $y(t)$ and $z(t)$ in a 2×2 layout. Symbols follow Figure 7.

The M4 initial speed is 32.05 m/s, exceeding the two placement values of 29.20 and 28.34 m/s. Its elevation is 10.46° and initial vertical velocity 5.82 m/s, producing a higher flight. As shown in Figure 10 and Table 3, M1 and M2 do not adequately capture the lateral bending. The M3 and M4 trajectories show how the upward initial velocity raises the ball, while gravity and the downward Magnus component in M4 bend the flight back. Magnus force therefore plays a key role in the lateral curvature of this power shot.

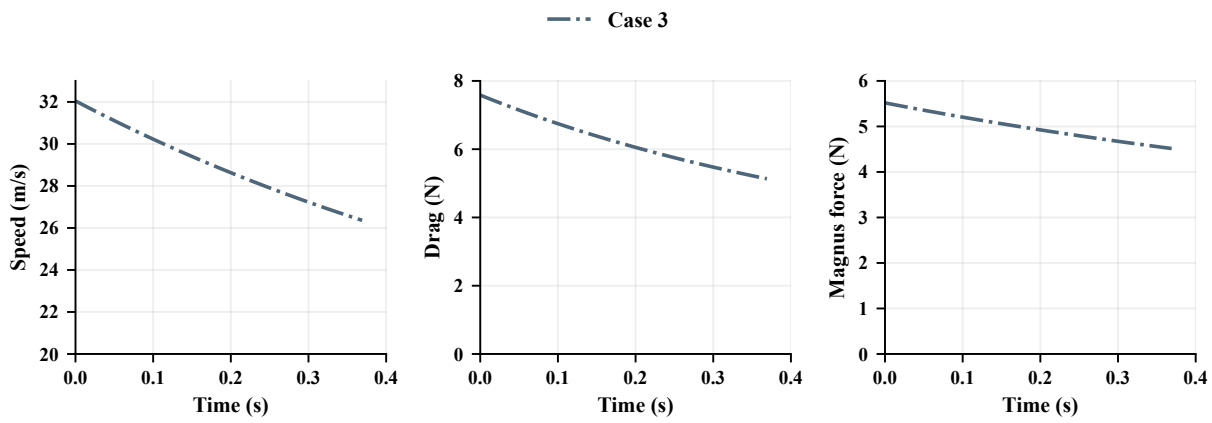


Figure 10: M4 speed, drag and Magnus-force magnitude for the power shot, on the same axes as Figure 8.

Table 3: Power-shot goal crossings and distance errors (m).

Case	Model	Observed (y, z)	Predicted (y, z)	Distance
Case 3	M1	$(2.919, 1.179)$	$(2.350, 1.558)$	0.684
Case 3	M2	$(2.919, 1.179)$	$(2.349, 1.506)$	0.657
Case 3	M3	$(2.919, 1.179)$	$(3.289, 1.126)$	0.374
Case 3	M4	$(2.919, 1.179)$	$(2.934, 1.291)$	0.113

Considering both a valid goal-plane prediction and its distance error, accounting for drag and spin, especially the changing Magnus force during flight, makes **M4 the better model for these three cases.**

4.3 M4 Inversion for the Problem B Observations

The initial centre is $(0, 0, 0.11)$ m. As shown in Table 4, Problem B specifies two later positions at 0.05 and 0.10 s. We apply the same M4 inverse method and parameter settings to these inputs.

Table 4: Initial position and two observations specified in Problem B.

t (s)	x (m)	y (m)	z (m)
0	0	0	0.11
0.05	1.50	0.45	0.44
0.10	2.96	0.86	0.74

The inferred initial motion is

$$\begin{aligned} \boldsymbol{v}_0 &= (30.3784, 9.4074, 6.8974)\ \mathrm{m/s}, \\ \boldsymbol{\omega}_0 &= (19.7434, 1.2060, -22.5553)\ \mathrm{rad/s}. \end{aligned} \tag{10}$$

As shown in Figure 11, **the predicted goal-plane crossing is** $(y_g, z_g) = (2.6159, 1.9526)$ **m**. The inferred speed of 32.54 m/s, elevation of 12.24° and pronounced lateral bending indicate that **this trajectory more closely resembles the power-shot type studied in this chapter.**

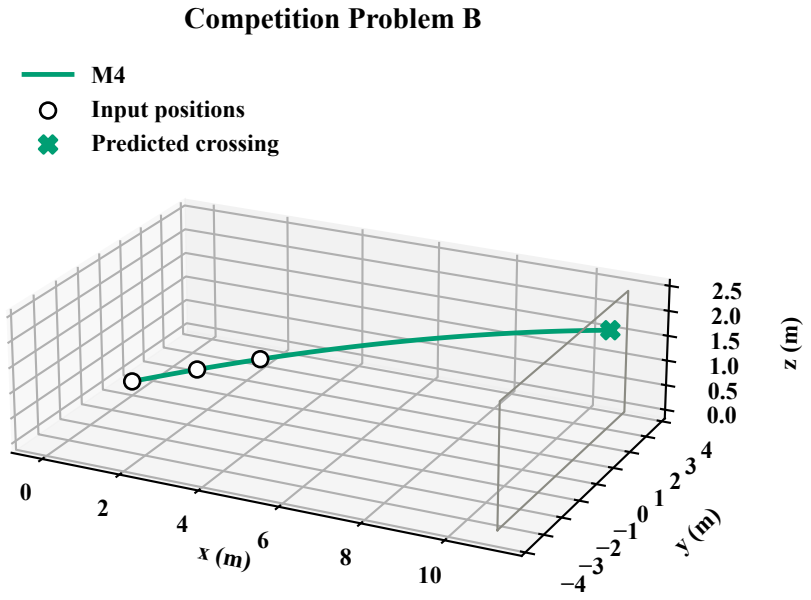


Figure 11: M4 prediction for Problem B, showing the three input positions, goal and predicted crossing.

Section 5 uses this inferred state to study changes in the goal crossing and its distribution.

5 Sensitivity Analysis

This chapter uses the M4 state inferred in Section 4.3 to examine how initial motion, aerodynamic parameters and air density affect the goal-plane crossing and arrival time. The model includes gravity, drag and Magnus force, with still air, constant spin, a fixed mass of 0.43 kg and a fixed radius of 0.11 m. The early positions in Table 4 also constrain the alternative trajectories considered below.

The reference crossing is $(y_g^*, z_g^*) = (2.6159, 1.9526)$ m, with arrival time $t_g^* = 0.398967$ s. The corresponding coefficients are $C_D = 0.3623$ and $\alpha = 2.2877$, and the maximum early-coordinate residual is 0.784 mm.

We first vary one parameter at a time to identify its effect on the crossing and arrival time, then retain the early-position constraints to compare predictions from different parameter combinations. Finally, joint perturbations generate the crossing distribution and its statistical ranges.

5.1 Effects of Individual Parameters on Crossing Position and Arrival Time

Each parameter is varied separately while the others remain at their reference values. Table 5 lists the nine perturbation ranges: percentages denote changes relative to the reference value, and direction angles vary about their reference orientation. Each parameter takes 41 equally spaced values, giving 369 trajectories.

Table 5: Perturbation ranges for the nine parameters

Parameter	Perturbation range
Launch speed	$\pm 5\%$
Velocity azimuth	$\pm 2^\circ$
Velocity elevation	$\pm 2^\circ$
Spin magnitude	$\pm 10\%$
Spin-axis azimuth	$\pm 2^\circ$
Spin-axis elevation	$\pm 2^\circ$
Drag coefficient	$\pm 10\%$
Lift parameter	$\pm 10\%$
Air density	$\pm 10\%$

The changes in crossing coordinates and arrival time relative to the reference trajectory are defined as

$$\begin{aligned} &\Delta y_g = y_g - y_g^*, \qquad \Delta z_g = z_g - z_g^*, \qquad \Delta t_g = t_g - t_g^*, \\ &D_g = \sqrt{(\Delta y_g)^2 + (\Delta z_g)^2}. \end{aligned} \tag{11}$$

Figures 12 and 13 show the crossing-position and arrival-time responses. Table 6 takes direction perturbations of $\pm 1^\circ$ and other perturbations of $\pm 5\%$ from these scans, reporting the larger displacement and absolute time change from each positive–negative pair.

Table 6: One-parameter effects on goal-plane crossing and arrival time

Parameter	Perturbation	Maximum D_g (cm)	Maximum $\lvert\Delta t_g\rvert$ (ms)
Launch speed	$\pm 5\%$	8.356	20.659
Velocity azimuth	$\pm 1^\circ$	20.070	1.965
Velocity elevation	$\pm 1^\circ$	20.218	1.692
Spin magnitude	$\pm 5\%$	3.873	0.287
Spin-axis azimuth	$\pm 1^\circ$	1.002	0.071
Spin-axis elevation	$\pm 1^\circ$	0.852	0.084
Drag coefficient	$\pm 5\%$	0.620	2.274
Lift parameter	$\pm 5\%$	3.873	0.287
Air density	$\pm 5\%$	4.142	1.990

For the perturbation sizes in Table 6, initial direction primarily affects crossing position: a 1° change in azimuth or elevation moves the crossing by about 0.20 m. Speed strongly affects arrival time; a $\pm 5\%$ change produces a maximum time difference of 20.659 ms and a maximum crossing displacement of 8.356 cm. Direction directly determines lateral and vertical motion, while speed changes the time to the goal and therefore the accumulated effects of gravity and aerodynamic forces.

The responses to spin magnitude and the lift parameter coincide because the linear Magnus force is proportional to the product $\alpha\boldsymbol{\omega}$. The same relative change in either spin magnitude or α produces the same force and trajectory response. Scaling them inversely gives

$$\boldsymbol{\omega}_0' = k\boldsymbol{\omega}_0, \qquad \alpha' = \alpha/k \quad \Longrightarrow \quad \boldsymbol{F}_L' = \boldsymbol{F}_L, \qquad k > 0. \tag{12}$$

Different spin magnitudes and lift parameters can therefore produce the same trajectory. Spin-axis orientation affects the magnitude and direction of deflection through $\boldsymbol{\omega} \times \boldsymbol{v}$. Air density scales both drag and Magnus force, changing deceleration and curvature together.

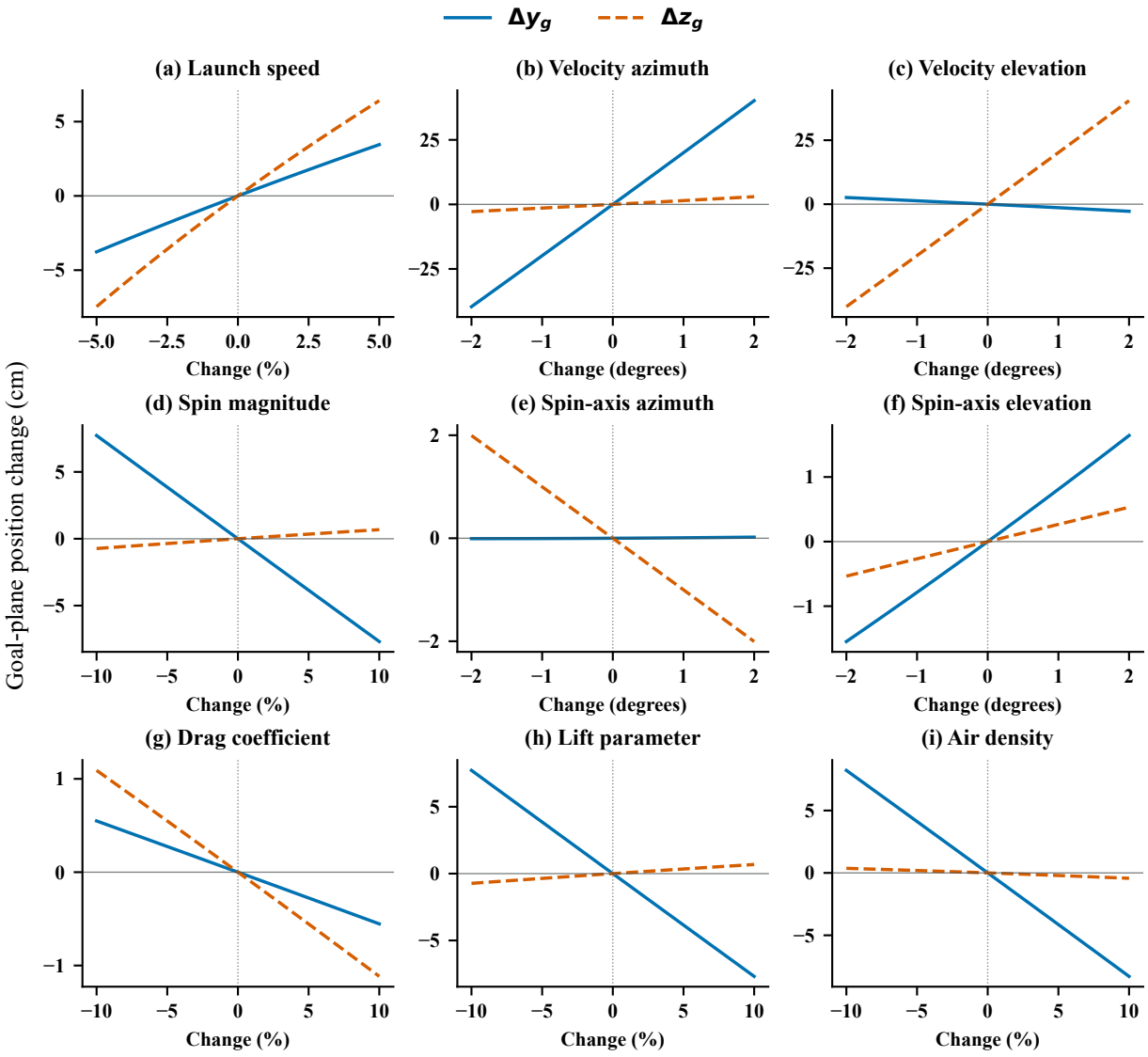


Figure 12: Goal-plane coordinate responses to nine one-parameter perturbations. Blue solid curves show Δy_g and orange dashed curves show Δz_g, in cm. Vertical scales differ between panels.

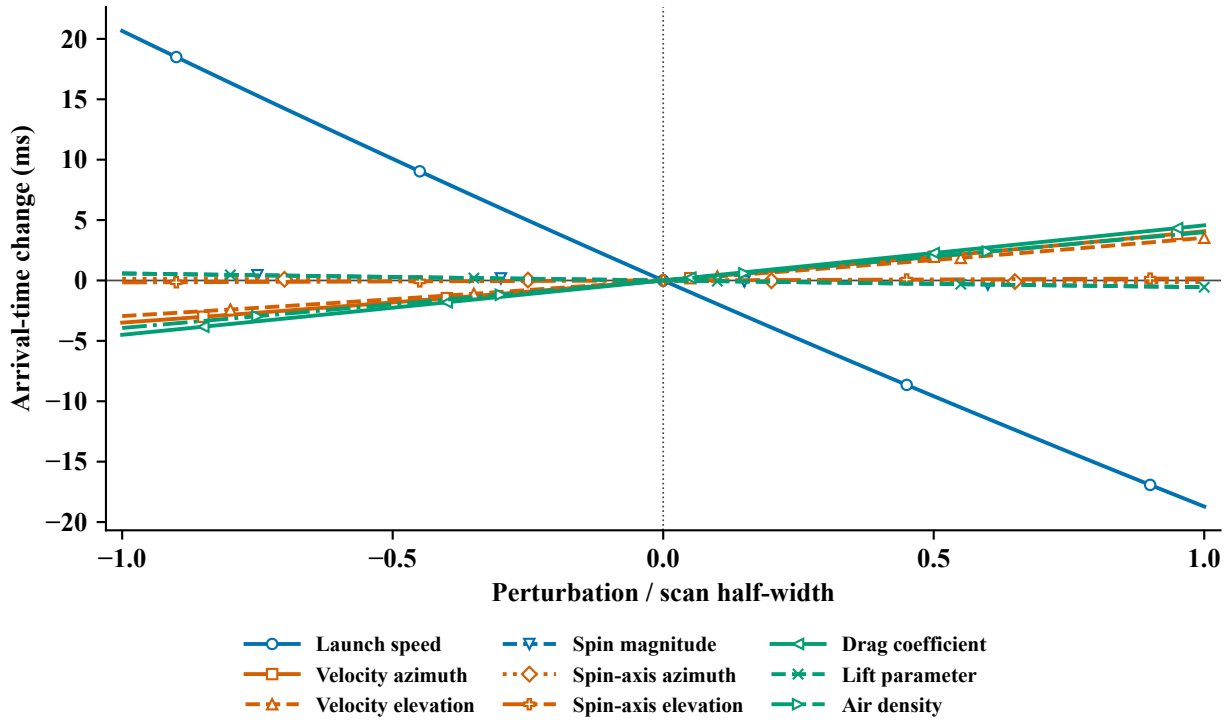


Figure 13: Arrival-time responses. Perturbation divided by scan half-width is plotted horizontally, so ± 1 denotes each scan endpoint. The vertical axis is Δt_g (ms); positive values mean later arrival.

5.2 Prediction Differences under Early-Position Constraints

The preceding analysis varies one parameter while holding the others fixed. We now examine how much the goal-plane crossing can change when different parameter combinations reproduce the same early positions approximately.

Using the aerodynamic reference values C_D^* and α^* from Section 4, define

$$\lambda_D = C_D/C_D^*, \qquad \lambda_\alpha = \alpha/\alpha^*. \tag{13}$$

Each coefficient multiplier takes 17 equally spaced values in $[1.2, 1.8]$, forming 17×17 combinations. For each combination, the initial spin component ω_{x0} takes 33 equally spaced values in $[-60, 60]$ rad/s. The three initial velocity components and the other two spin components are then fitted to the early positions in Table 4. The constraints are a total speed not exceeding 35 m/s, an absolute value not exceeding 60 rad/s for each spin component, and a maximum early-coordinate residual not exceeding 2 mm.

The supplementary search yields 5049 compatible trajectories across 153 aerodynamic-parameter combinations. Restricting both multipliers to $[1.35, 1.65]$, corresponding to $\pm 10\%$ changes in each aerodynamic parameter about its reference-trajectory value, gives 2079 trajectories across 63 combinations. Table 7 lists their crossing and arrival-time ranges, and Figure 14 shows their crossing positions.

Table 7: Goal-plane crossing ranges under early-position constraints

Range of each multiplier	y_g (m)	z_g (m)	t_g (s)
$[1.35, 1.65]$	2.5538–2.6550	1.9061–2.0369	0.3972–0.4025
$[1.2, 1.8]$	2.5478–2.6601	1.8972–2.0430	0.3972–0.4043

Within the wider parameter range, the largest crossing displacement among the searched trajectories is 0.1074 m relative to the reference. The minimum-residual solutions for each aerodynamic-parameter combination lie at the spin-component bounds; plotting these solutions alone produces two clusters. Intermediate spin states satisfying the same 2 mm position constraint form the band joining the clusters in Figure 14. Thus, initial velocity and spin can compensate for aerodynamic changes over the early observation interval, while different force histories accumulate into distinct subsequent crossings.

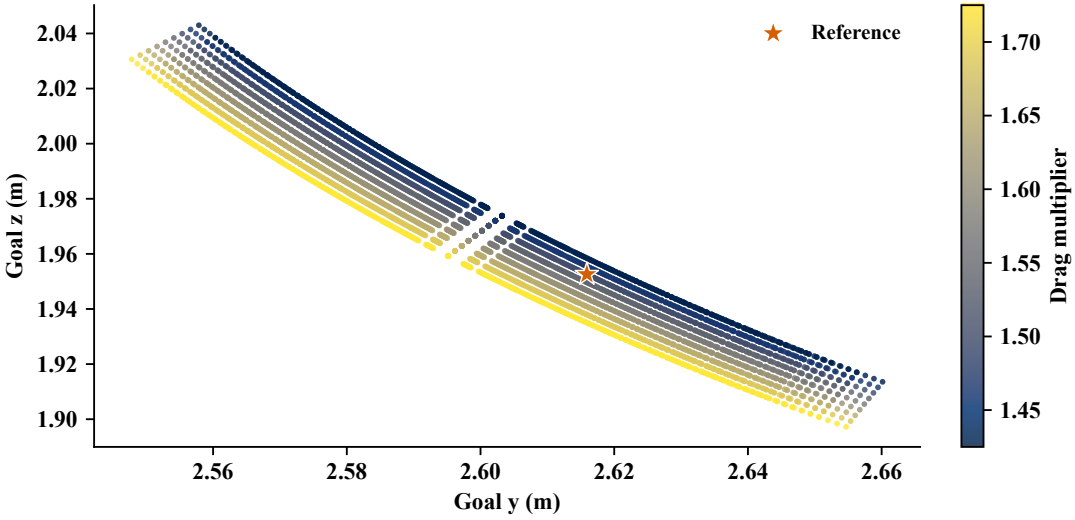


Figure 14: The 5049 crossings from the supplementary search under the 2 mm early-position constraint. Colour denotes λ_D; the orange star marks the reference trajectory.

5.3 Crossing Distribution under Joint Perturbations

To examine simultaneous changes in several conditions, the nine parameters are sampled independently and uniformly over the ranges in Section 5.1, producing 100000 inputs. Mass, radius and initial position remain fixed. Integrating the equations for each input gives the crossing distribution under jointly varying launch conditions and aerodynamic parameters.

Table 8: Crossings and arrival times for 100000 jointly perturbed trajectories

Output	Sample minimum–maximum	Central 95% interval, one coordinate
y_g (m)	1.9633–3.2143	2.1858–3.0505
z_g (m)	1.4381–2.4646	1.5534–2.3535
t_g (s)	0.3685–0.4346	0.3783–0.4218

The central 95% intervals in Table 8 use the 2.5th and 97.5th percentiles of each output. The mean crossing is $(2.6170, 1.9511)$ m, with lateral and vertical standard deviations of 0.2455 m and 0.2360 m. The largest displacement from the reference crossing is 0.801 m.

To describe lateral and vertical variation jointly, write $\boldsymbol{q}_i = (y_{g,i}, z_{g,i})^{\mathsf{T}}$ and use the sample mean $\bar{\boldsymbol{q}}$ and covariance matrix $\widehat{\Sigma}$ to define

$$d_i^2 = (\boldsymbol{q}_i - \bar{\boldsymbol{q}})^{\mathsf{T}} \widehat{\Sigma}^{-1} (\boldsymbol{q}_i - \bar{\boldsymbol{q}}). \tag{14}$$

Taking the empirical 95th percentile $c_{0.95} = 4.5916$ of d_i^2 gives the joint region

$$\mathcal{E}_{0.95} = \left\{ \boldsymbol{q} : (\boldsymbol{q} - \bar{\boldsymbol{q}})^{\mathsf{T}} \widehat{\Sigma}^{-1} (\boldsymbol{q} - \bar{\boldsymbol{q}}) \le c_{0.95} \right\}. \tag{15}$$

The set $\mathcal{E}_{0.95}$ is the ellipse boundary and its interior in Figure 15. The ellipse line marks the boundary, and the region contains 95% of the simulated crossings. The goal is 2.44 m high; accounting for the 0.11 m ball radius requires a centre height between 0.11 and 2.33 m and a lateral position satisfying $|y_g| \le 3.55$ m for the whole ball to pass through. By these criteria, 95772 trajectories clear the opening, or 95.772%; the other 4228 trajectories, or 4.228%, exceed the 2.33 m centre-height limit.

Figure 16 displays 400 randomly selected trajectories, linking the nine inputs to crossing coordinates and arrival time. Inputs are scaled to 0–1 using their perturbation bounds; outputs use the extrema of the full 100000-trajectory sample. The subset contains 385 blue lines for trajectories clearing the goal and 15 orange lines for trajectories crossing too high. Each line connects one set of launch conditions and aerodynamic parameters to its predicted outcome.

Figure 17 checks whether the crossing statistics stabilize as the number of simulations increases, informing the choice of 100000 samples for estimating the central 95% intervals. The blue sample extrema expand as additional extreme trajectories appear, while the orange percentile intervals gradually stabilize. Increasing the sample from 50000 to 100000 changes the lateral and vertical interval endpoints by at most 0.264 mm and 0.506 mm, and the arrival-time endpoints by at most 0.138 ms. Two additional runs of 10000 trajectories with independent random seeds differ from the main sample by at most 5.425 mm laterally, 4.320 mm vertically and 0.328 ms in time. These comparisons show that further sampling has a small effect on the central-interval estimates.

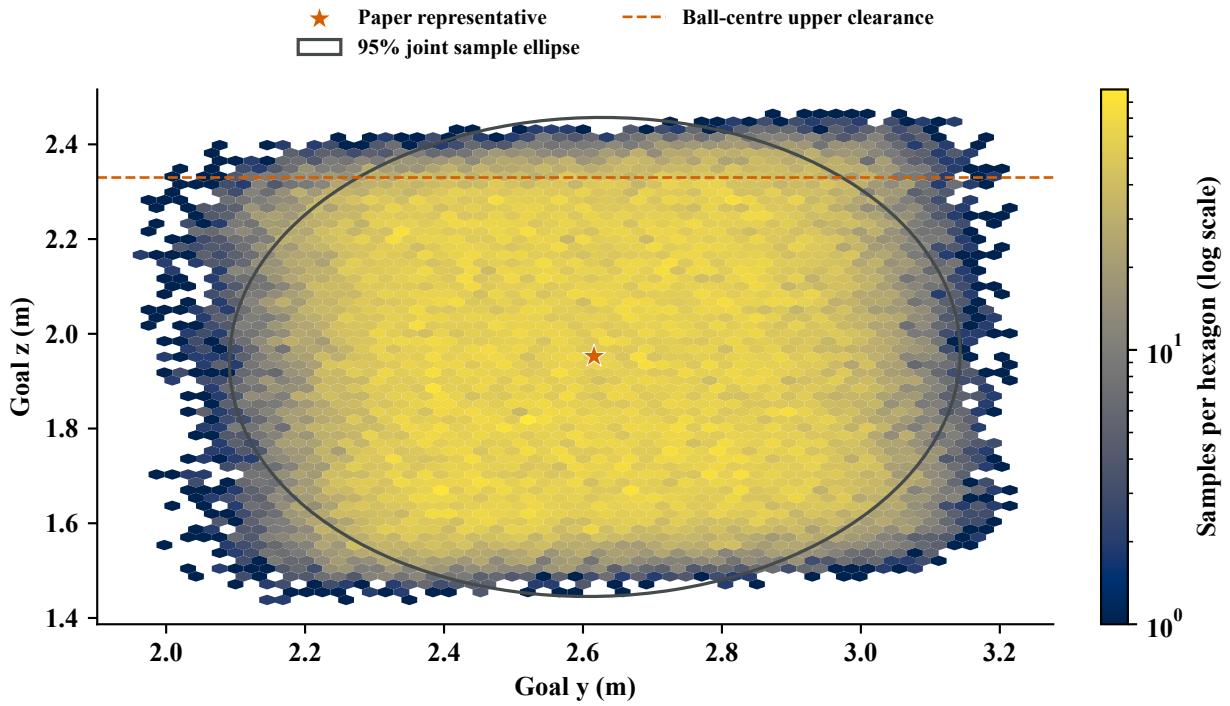


Figure 15: Goal-plane crossings under 100000 joint perturbations. Hexagon colour indicates sample count, the ellipse denotes the joint 95% region, and the dashed line marks the ball-centre height limit $z_g = 2.33$ m.

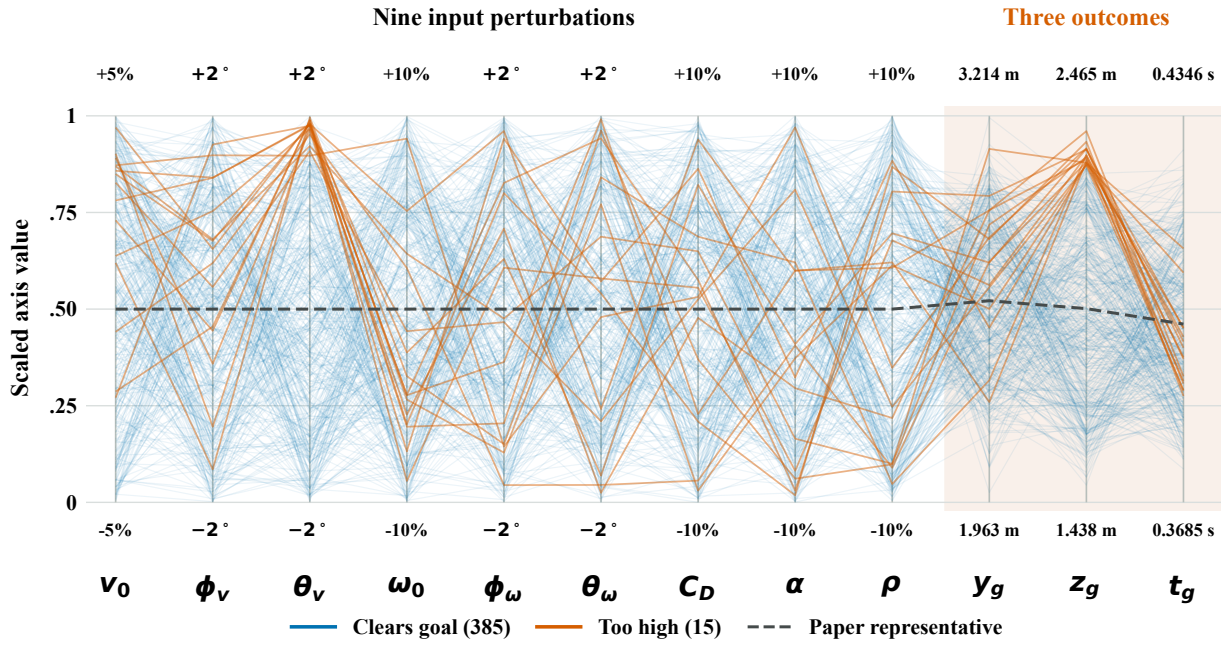


Figure 16: Parallel coordinates linking perturbed inputs, crossing coordinates and arrival time. The first nine axes are inputs and the last three are outputs. The displayed subset contains 385 trajectories clearing the opening and 15 crossing too high.

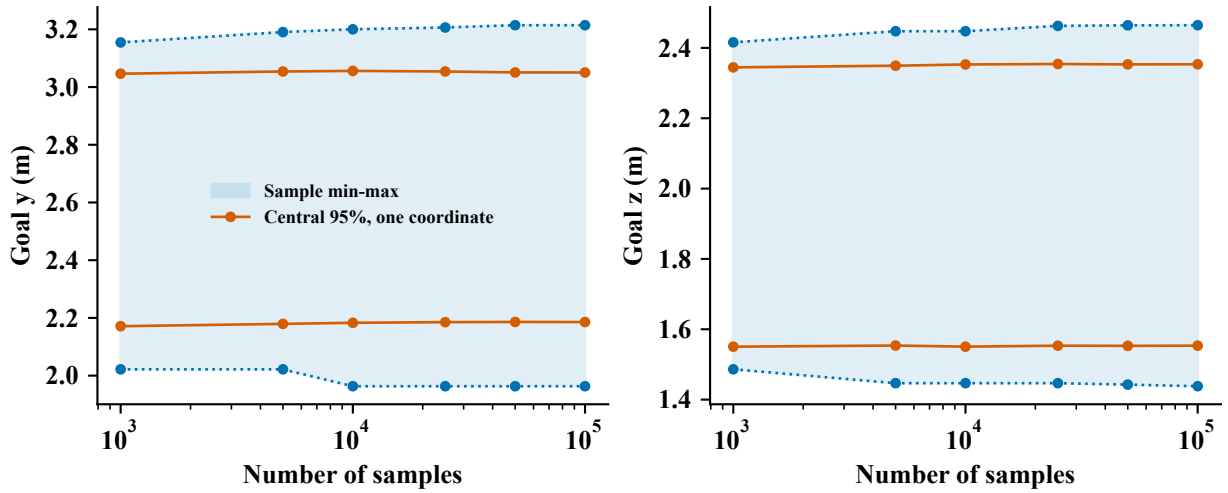


Figure 17: Convergence of crossing statistics with simulation sample size. Blue shading and dotted curves show sample extrema; orange curves show each coordinate's central 95% interval, illustrating the stability of interval estimates as sample size increases.

6 Discussion of Other Factors

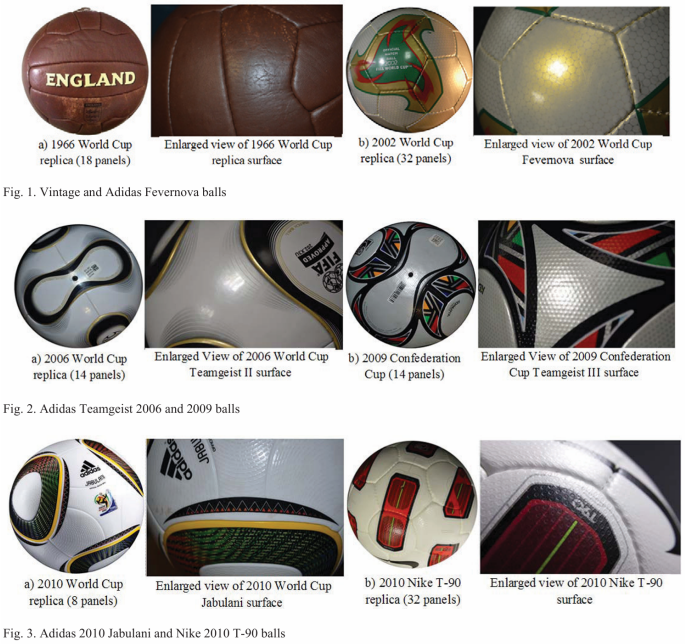
This chapter examines the rationale for neglecting geometric details, spin evolution and air motion in the Chapter 3 model.

6.1 Asymmetric Football Surface Geometry

As shown in Figure 18, football seams, panels and textures differ from an ideal spherical surface. We refer to the geometric descriptions of seamed footballs and smooth spheres in Paudel and Yan [4]; the corresponding geometry, panels and textures appear in Figures 18a and 18b.



(a) Illustrative comparison of a seamed football and a smooth sphere.



(b) Six footballs and surface details, reproduced from Figures 1–3 of Alam et al. [10].

Figure 18: Football geometry, panels and surface textures.

According to the wind-tunnel measurements of Alam et al. [10], as shown in Figure 19, the trajectory in Section 4.3 spans Reynolds numbers $(3.75\text{–}4.77) \times 10^5$, over which the football drag coefficient varies relatively little. The additional geometric correction from resolving seams and textures is therefore small for this short-range centre-of-mass prediction. A spherical envelope is appropriate for describing the motion and projected area.

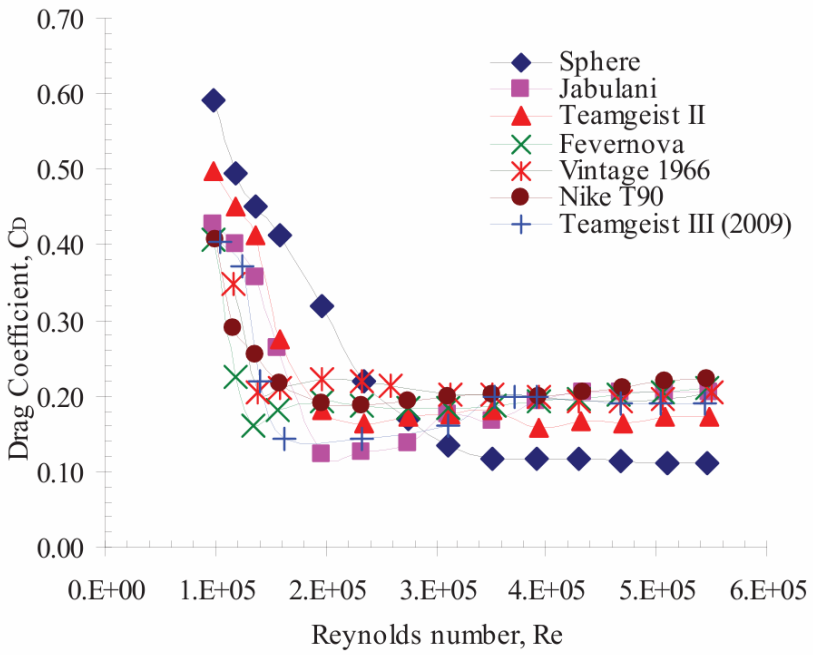


Figure 19: Drag coefficient against Reynolds number for different footballs and a smooth sphere, reproduced from Figure 5 of Alam et al. [10].

6.2 Rotation and Aerodynamic Damping

Chapter 3 neglects aerodynamic torque and the resulting change in angular velocity. According to the aerodynamic treatment of Grad et al. and MuJoCo's high-Reynolds-number model [7, 8], rotational damping is the principal torque responsible for spin decay during free flight. We now include this effect. For a sphere of radius R, the damping torque is

$$\boldsymbol{Q} = -\frac{8\pi}{15}\rho C_{\text{angular}} R^5 \|\boldsymbol{\omega}\|\boldsymbol{\omega}. \tag{16}$$

Here C_{angular} is the dimensionless angular-drag parameter, fixed at $C_{\text{angular}} = 1.5$ following MuJoCo [8].

Following the thin-shell approximation of Maponi and Piombin [3], the inertia is $I_b = \frac{2}{3}mR^2$. Incorporating Eq. (16) into the Chapter 3 motion equation (7), with the forces in Eqs. (3) and (4), leads to

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ I_b\dot{\boldsymbol{\omega}} = \boldsymbol{Q}. \end{cases} \tag{17}$$

The initial conditions follow Eq. (8). Using Eq. (17), we repeat the inference with the same starting position and observations at 0.05 and 0.10 s as Section 4.3, obtaining the following comparison.

As shown in Figure 20, the fixed rotational damping gradually reduces spin to 25.6921 rad/s at the goal, a decrease of 14.36% from its initial value. As shown in Table 9, after refitting, the two goal crossings differ by 2.38 cm.

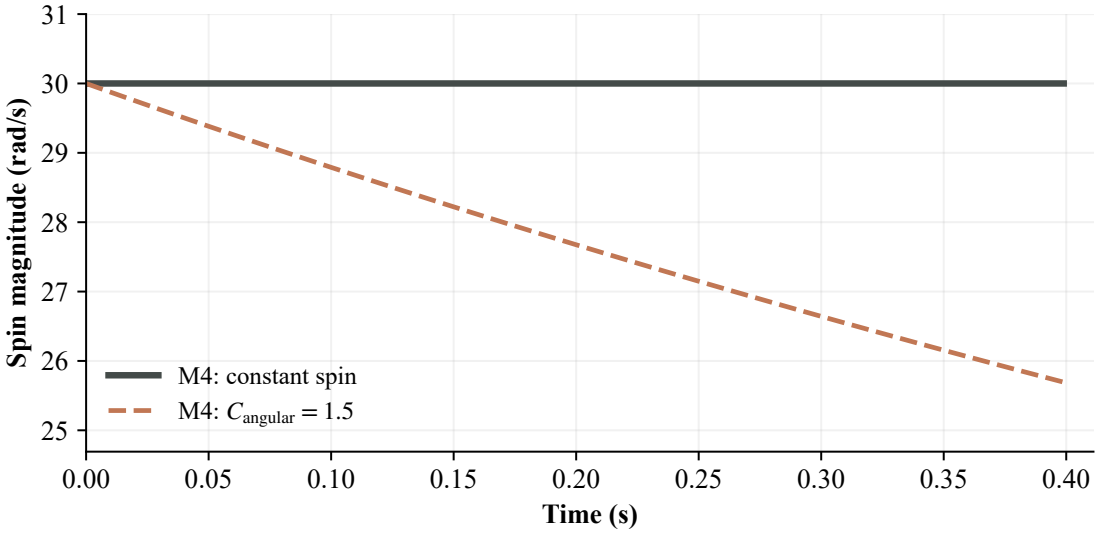


Figure 20: Angular velocity as a function of time.

Table 9: Goal crossing and spin; the damping model fixes $C_{\text{angular}} = 1.5$.

Model	y_g (m)	z_g (m)	t_g (s)	$\|\boldsymbol{\omega}_g\|$ (rad/s)
M4	2.6159	1.9526	0.398967	30.0000
M4 + torque	2.6397	1.9515	0.399139	25.6921

Rotational damping weakens spin and hence Magnus force, gradually reducing the subsequent curvature. Over the approximately 0.4 s flight considered here, this change accumulates to a centimetre-scale difference at the goal. The constant-spin approximation therefore retains good endpoint prediction for this short flight.

6.3 A Computed Wind-Speed Limit

Chapter 3 also neglects wind. We now examine the range of wind speeds for which its effect on the prediction is small. Following the relative-airflow treatment of Maponi and Piombin [3], let air velocity be $\boldsymbol{v}_{\text{wind}}$ and air-relative velocity be $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$. Thus $\boldsymbol{u}$ replaces $\boldsymbol{v}$ in the Chapter 3 aerodynamic forces. With $q_w = \frac{1}{2}\rho u^2$ and $q_0 = \frac{1}{2}\rho v^2$ denoting dynamic pressure with wind and in still air, respectively,

$$\frac{|q_w - q_0|}{q_0} \leq 2\frac{w}{v} + \left(\frac{w}{v}\right)^2. \tag{18}$$

From Eq. (18), drag scales quadratically with relative speed, while linear Magnus force varies with $\boldsymbol{\omega} \times \boldsymbol{u}$. The wind-to-ball speed ratio therefore describes the force change under weak wind.

Using the M4 initial state and aerodynamic parameters of Section 4.3 as the reference, we scan the wind-speed limits for a goal-crossing displacement of no more than 5 cm.

Table 10: Wind-speed limits for a goal-crossing shift of at most 5 cm.

Wind direction	Speed limit (m/s)
Tailwind	1.79
Headwind	1.69
Left crosswind	1.13
Right crosswind	1.15
All horizontal directions	0.94

As shown in Table 10, the representative directions have corresponding wind-speed limits. Over the horizontal-direction scan, $w \leq 0.94$ m/s produces a maximum computed crossing displacement of 4.953 cm, within 5 cm. The still-air approximation therefore meets the endpoint-deviation criterion in this weak-wind range.

7 Conclusions

The four models describe gravitational fall, deceleration, a constant resultant force and rotational deflection. M3 obtains constant acceleration from three positions. M4 inserts constant C_D and $C_L = \alpha S p$ into the motion equations, updating drag and Magnus force as speed changes.

For M4, the crossing errors are 0.360, 0.146 and 0.113 m for the three penalties. Its power-shot crossing error and subsequent path RMSE are the smallest among the four models. For Case 1, M3 passes through the early positions but reaches a maximum $x = 10.973$ m, short of the goal; M4 supplies a goal-plane prediction for this low flight.

In the competition-position example, a 1° launch-direction change moves the crossing by about 0.20 m, while speed strongly affects arrival time. Spin magnitude and α compensate through the linear lift law. The largest grid-search shift under the 2 mm position constraint is 0.1074 m; 95.772% of 100,000 jointly perturbed flights clear the opening.

With $C_{\text{angular}} = 1.5$ fixed and the competition positions refitted, spin decreases by 14.36% at the goal and the crossing moves 2.38 cm relative to M4. For horizontal winds up to 0.94 m/s, the direction scan keeps the crossing shift within 5 cm. These calculations support the constant-spin and still-air approximations for this problem.

8 Model Evaluation

8.1 Model Strengths

Connecting forces and trajectory changes. The model incorporates gravitational fall, drag-induced deceleration and rotational deflection in a common equation of motion, relating trajectory curvature to changes in force magnitude and direction. The four-model comparison shows how different force descriptions and the constant-acceleration approximation affect predictions, providing a physical interpretation of the differences between low placement shots and higher power shots.

Connecting early inference and later validation. Three early positions are used to estimate initial motion and relevant parameters, after which the corresponding equations predict the subsequent trajectory. Later measured positions provide comparisons of the path and goal-plane crossing, connecting early fitting with subsequent predictive validation. Shared coordinates and definitions of ground contact and goal-plane crossing also make results comparable across models and shot cases.

Reference predictions and crossing ranges. Individual-parameter scans distinguish the effects on crossing position and arrival time. The position-constrained search reveals differences in predictions from parameter combinations matching the early observations, while joint perturbations give crossing distributions over specified ranges. Together, these analyses connect parameter changes to the final prediction, describing a reference crossing, response trends and distribution ranges.

8.2 Model Weaknesses

Coupling of initial motion and aerodynamic parameters. Over a short observation interval, initial velocity, spin and aerodynamic parameters can compensate for one another, allowing initially similar trajectories to separate later in flight. In the linear lift relation, spin magnitude and the lift parameter enter the force through their product, so different parameter values can also generate the same trajectory. Additional early tracking positions and direct spin measurements would further constrain initial motion and aerodynamic parameters.

Simplified aerodynamics and environmental conditions. The main model treats the drag coefficient and linear lift parameter as constants during each flight and assumes still air and constant spin. Flow transitions, ball orientation and wake fluctuations alter aerodynamic forces, while wind and rotational damping affect relative airflow and spin evolution. More detailed descriptions require aerodynamic data for each ball type, together with wind and spin measurements.

Limited coverage of measured conditions. The measured comparison covers two placement shots and one power shot. Tracking records spanning additional ball types, shot techniques and wind conditions would allow assessment across more operating conditions. Joint perturbations use independent uniform sampling of nine parameters; estimating their distributions and correlations from measurements would bring the simulated crossing distribution closer to actual shooting conditions.

References

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Appendix A: Data Sources, Frame Selection and Parameter Bounds

Following the dataset description by Bassek et al. [1] and the DFL release [2], we retain the source event and frame identifiers. The three selected penalties contain 39 records, with 15, 12 and 12 for the respective events. As shown in Table 11, each case is associated with its player and selected samples.

Table 11: Correspondence between case labels and source records

Case	Player	Selected samples
Case 1	Dejan Ljubicic	3, 4, 5
Case 2	Philipp Klement	3, 4, 5
Case 3	Dawid Igor Kownacki	2, 3, 4

The first selected position is the time origin and the fitting times are 0, 0.04 and 0.08 s. Their source-time origins are 0.08, 0.08 and 0.04 s in the table's order. Repeated and post-goal records are retained; later positions are used only for validation. The observed goal crossing uses a source frame or interpolation. Let $(X, Y, Z_{\text{original}})$ denote the published pitch-centred coordinates. The height is interpreted in metres as ball-centre height, and the local coordinates follow from $x = X - 41.5$, $y = Y$ and $z = Z_{\text{original}}$.

As shown in Table 2, the retained inputs and observed crossings share one local coordinate system. As shown in Table 12, the inverse variables have specified search ranges.

Table 12: Component bounds and force-correction settings for inversion.

Quantity	Search interval
v_{x0} (M1, M2, M4)	$[0, 35]$ m/s
v_{y0}, v_{z0} (M1, M2, M4)	$[-35, 35]$ m/s
$\|\boldsymbol{\omega}_0\|$ (M4)	30 rad/s
Azimuth, elevation (M4)	$[-\pi, \pi], [-\frac{\pi}{2}, \frac{\pi}{2}]$ rad
$\frac{C_D}{C_D^*}, \frac{\alpha}{\alpha^*}$	$[0.5, 1.5]$

As shown in Table 13, the M4 states differ in speed, coefficients and unit spin axis $\boldsymbol{n} = \dfrac{\boldsymbol{\omega}}{\|\boldsymbol{\omega}\|}$.

Table 13: Fitted M4 states at the selected starting frames.

Event	v_0 (m/s)	C_D	α	n_x	n_y	n_z
Case 1	29.20	0.3623	2.2877	0.2280	-0.9728	0.0413
Case 2	28.34	0.1208	2.2877	0.1439	-0.2711	-0.9517
Case 3	32.05	0.3238	2.2877	-0.2276	0.3170	0.9207

Reproduction Code

The following excerpts evaluate the force components and state derivatives used in Eq. (7). Initial-state inference and subsequent trajectory prediction use these same physical relations and coefficient settings.

```
def state_derivative(t, state, config):
    """The single nine-state RHS used for prediction and inverse fitting."""
    f = forces(state[3:6], state[6:9], config)
    return np.r_[state[3:6], sum(f.values())/config.mass,
                 rotational_torque(state[6:9], config)/config.moment_of_inertia]
```

```
def forces(velocity, spin, config):
    """Return force components on ball, in newtons."""
```

```
    v = np.asarray(velocity, dtype=float) - np.asarray(config.wind, dtype=float)
    omega = np.asarray(spin, dtype=float)
    speed = np.linalg.norm(v)
    weight = (config.mass * np.asarray(config.constant_acceleration, dtype=float)
              if config.constant_acceleration is not None else
              np.array([0., 0., -config.mass * config.gravity if config.use_gravity else
                  0.]))
    drag = np.zeros(3)
    magnus = np.zeros(3)
    if speed > 1e-12:
        cd, cl, _, _, _ = coefficients(velocity, spin, config)
        q_area = 0.5 * config.air_density * np.pi * config.radius**2 * speed**2
        if config.use_drag:
            drag = -q_area * config.drag_scale * cd * v / speed
        cross = np.cross(omega, v)
        norm = np.linalg.norm(cross)
        if config.use_magnus and norm > 1e-12:
            # The literature fit is for transverse spin. Project its effect
            # onto the plane perpendicular to airflow for an oblique spin axis.
            sin_angle = norm / (np.linalg.norm(omega) * speed)
            magnus = q_area * config.lift_scale * cl * sin_angle * cross / norm
    return {"gravity": weight, "drag": drag, "magnus": magnus}
```